

AI-CliniScan: Milestone 1 Report

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Objective The primary objective of Milestone 1 (Weeks 1-2) is to acquire and prepare the VinDr-CXR dataset, set up the development environment, and create initial data processing pipelines. This ensures the raw clinical data is formatted correctly for downstream classification and detection model training.

Part 1: Dataset Description Theory The project relies on the VinDr-CXR dataset, containing 18,000 chest X-ray images with associated bounding boxes and diagnosis labels. The raw data is provided in DICOM format, with bounding box annotations stored in a CSV file. The data represents various pathological findings, such as opacities, consolidations, fibrosis, and masses.

Part 2: Data Preprocessing Theory Because deep learning models like YOLOv8 and ResNet cannot natively process raw DICOM files or mixed-format CSV annotations, a rigorous preprocessing pipeline was established.

- **DICOM to PNG Conversion:** Raw DICOM files are parsed to extract pixel arrays and converted into PNG format. During this step, the PhotometricInterpretation tag is evaluated to correct MONOCHROME1 images, ensuring all X-rays display a consistent standard of white bones on a dark background.
- **Intensity Normalization:** The pixel values undergo Min-Max normalization. This scales the widely varying raw pixel intensities strictly between 0 and 255, standardizing the brightness across images taken from different X-ray machines.
- **Resizing:** To meet the fixed input layer requirements of Convolutional Neural Networks, all images are resized to a uniform dimension.
- **Light Contrast Enhancement:** The Contrast Limited Adaptive Histogram Equalization (CLAHE) method is applied to the images. This enhances local contrast within the lung fields, highlighting subtle pulmonary abnormalities without overly amplifying background noise.
- **Annotation Conversion:** The Kaggle CSV bounding box coordinates are converted into the YOLO format. This involves mathematically transforming absolute bounding box coordinates (x_{min} , y_{min} , x_{max} , y_{max}) into normalized center coordinates (x_{center} , y_{center} , width, height) scaled between 0.0 and 1.0, and saving them as individual .txt files for each image.